

Advanced machine learning

Active learning

Transductive active learning: ITL and safe Bayesian optimization



Joachim Buhmann and Carlos Cotrini

Agenda

Model validation
and uncertainty
quantification

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ML models

Statistical estimators

Linear estimators

Neural networks

Transductive active
learning

Batch active learning

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Cross-validation



Bootstrap



Wald test



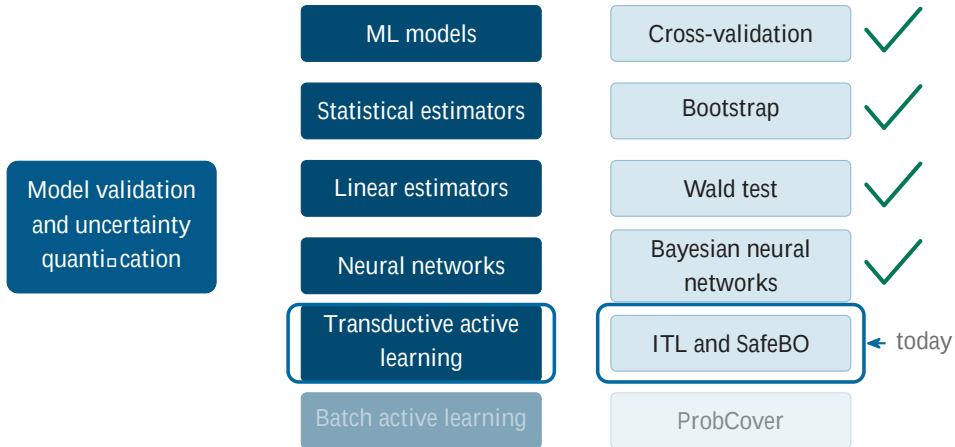
Bayesian neural
networks



ITL and SafeBO

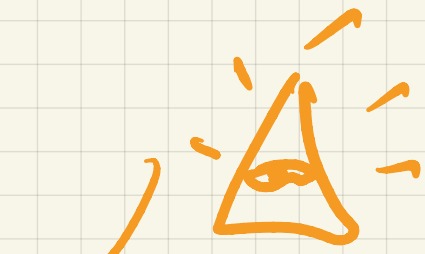
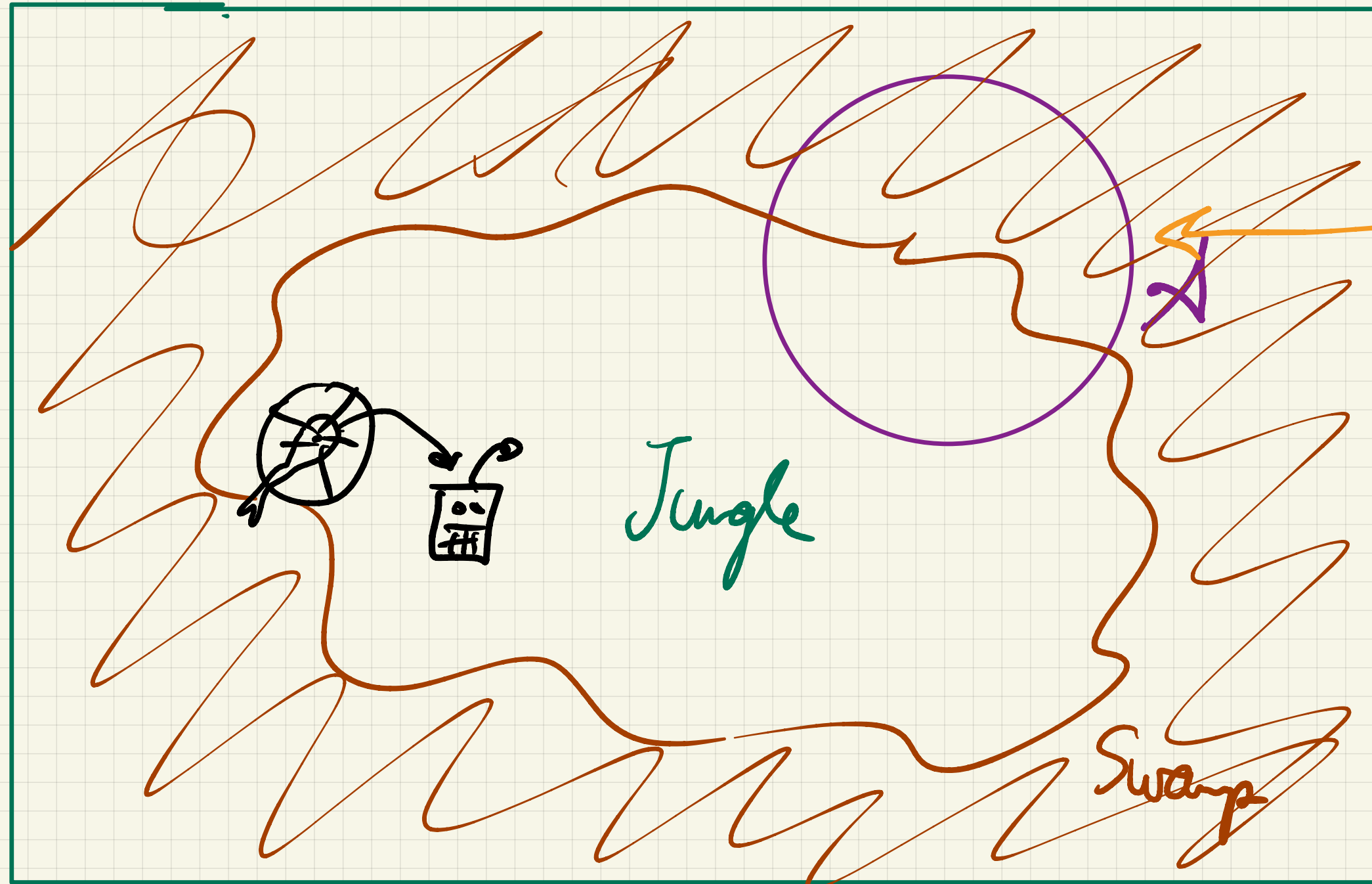
ProbCover

Agenda



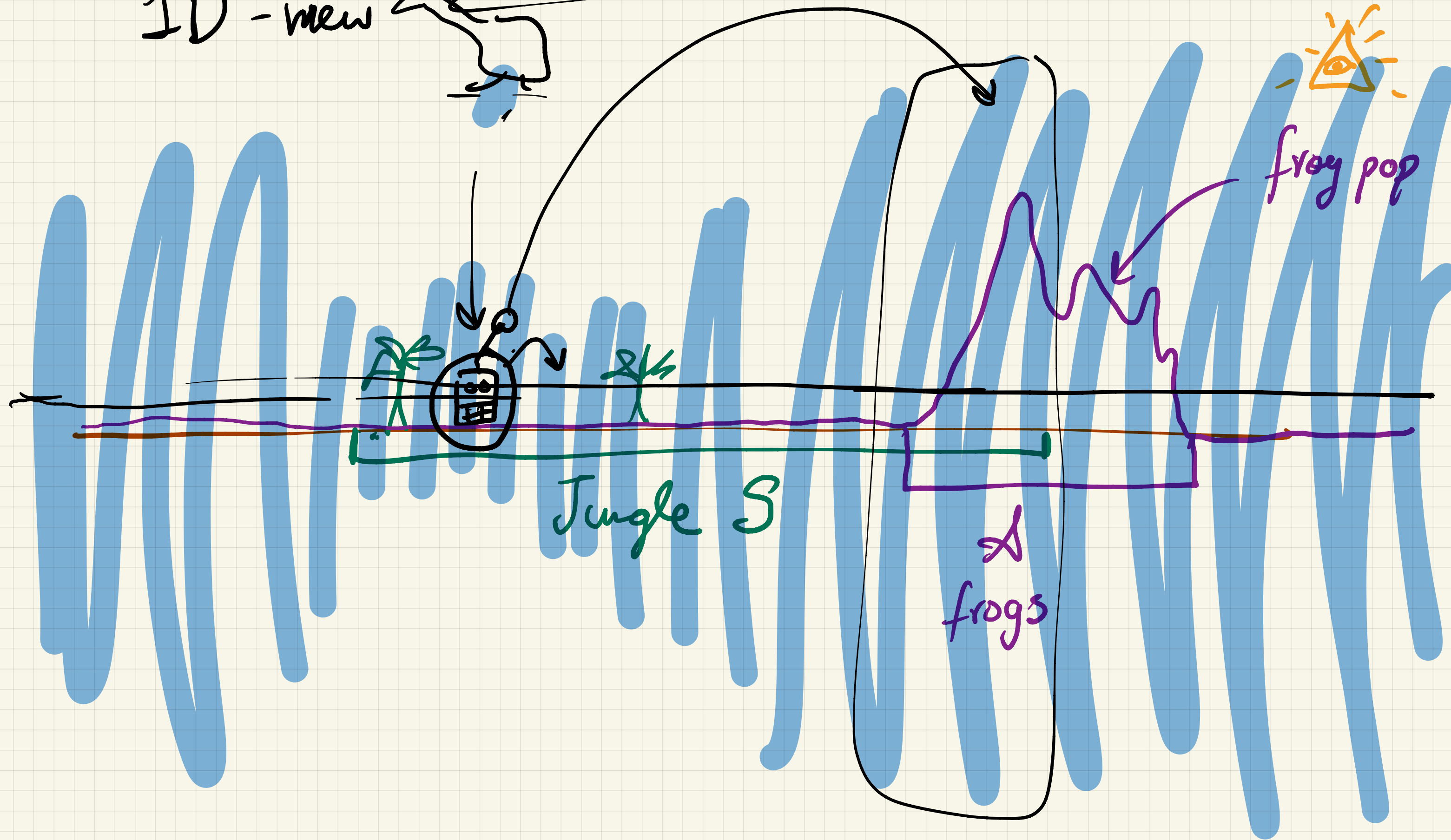
Active learning

1) GP



S

1D-view



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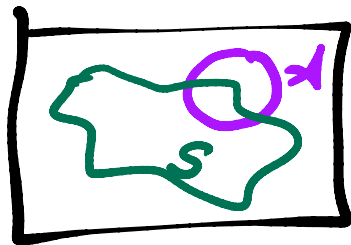
Active learning

- In some scenarios, it is hard to collect a representative sample of labeled data to train a model on it.
 - Diagnoses of rare diseases.
 - Autonomous driving.
 - Corporate fraud and data leakage.
- How can we collect a small yet representative sample when we are under a strict budget?



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- Assume a domain space $\mathcal{X}$, a target space $\mathcal{A} \subseteq \mathcal{X}$, and a sample space $\mathcal{S} \subseteq \mathcal{X}$.



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- Given $\mathcal{D}_{n-1} = \{(x_i, y_i)\}_{i < n} \subseteq \mathcal{S} \times \mathbb{R}$, find $x_n \in \mathcal{S}$ giving the most information about f in $\mathcal{A}$.

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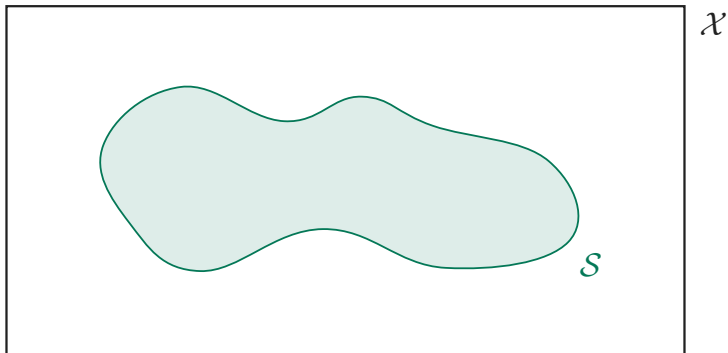
Key questions.

- How much information does a point (x_n, y_n) provide about f ?
- How do we select $x_n \in \mathcal{S}$ to gain the most about f in $\mathcal{A}$?

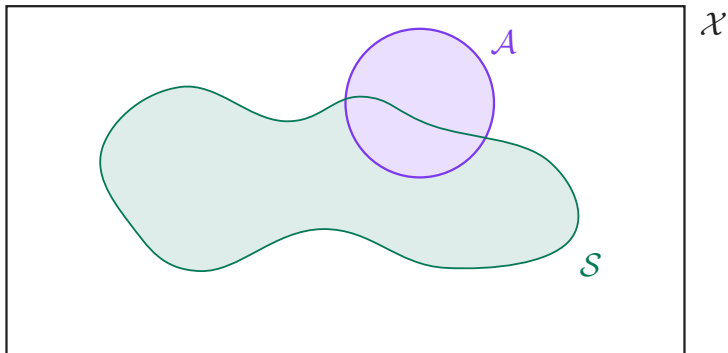
The goal, geometrically



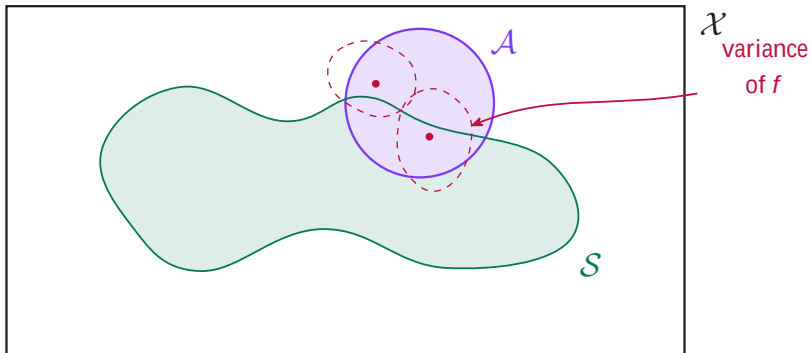
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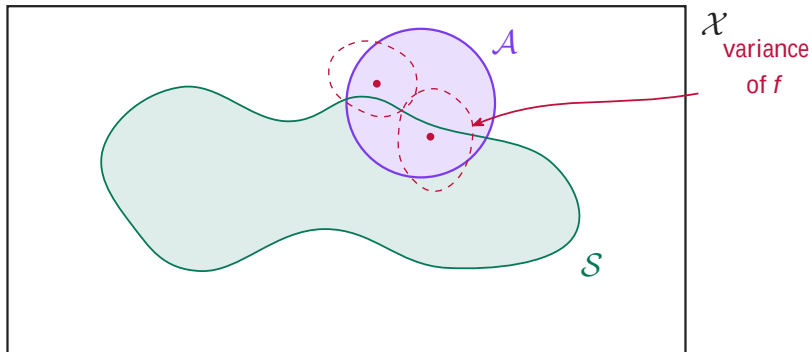
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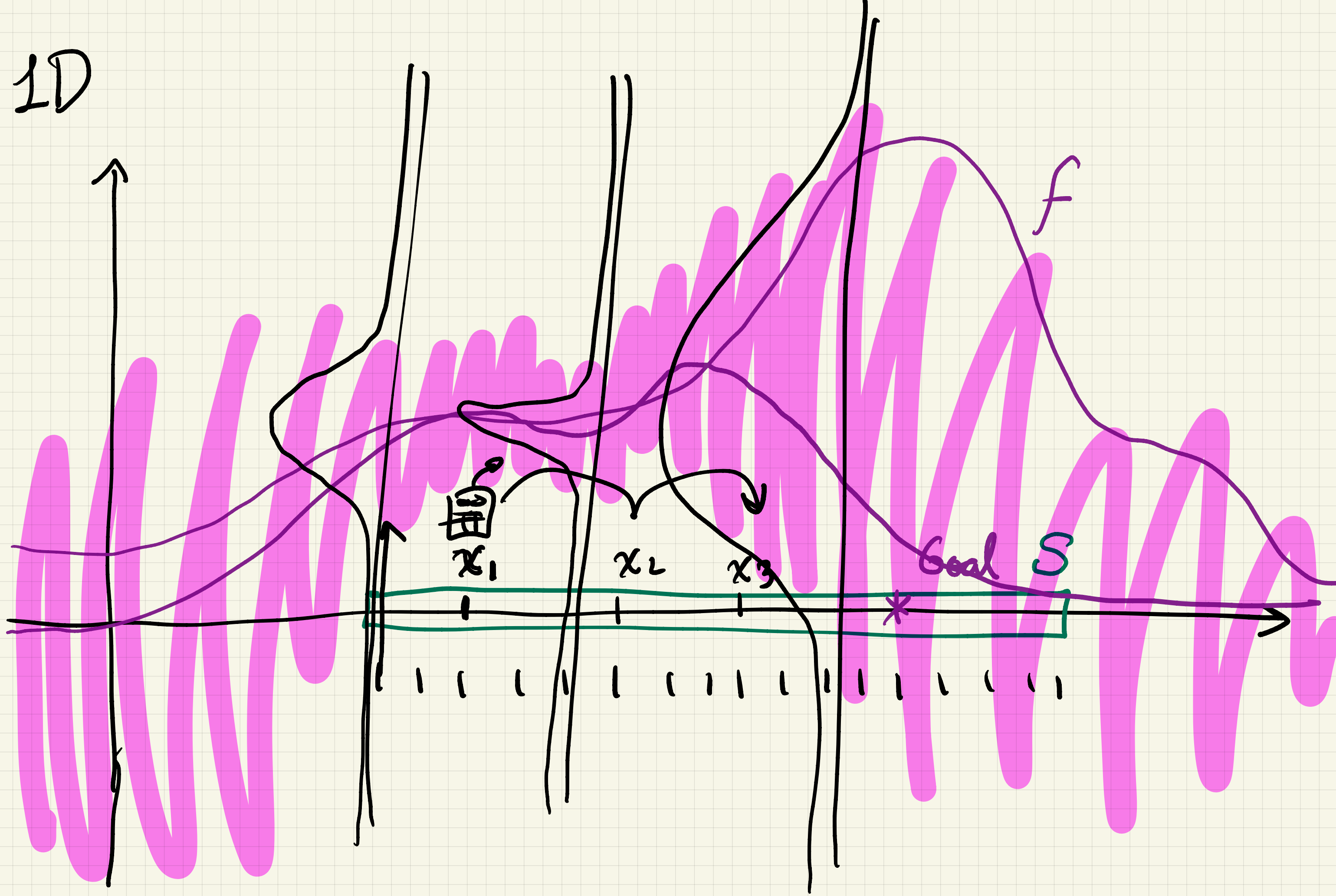
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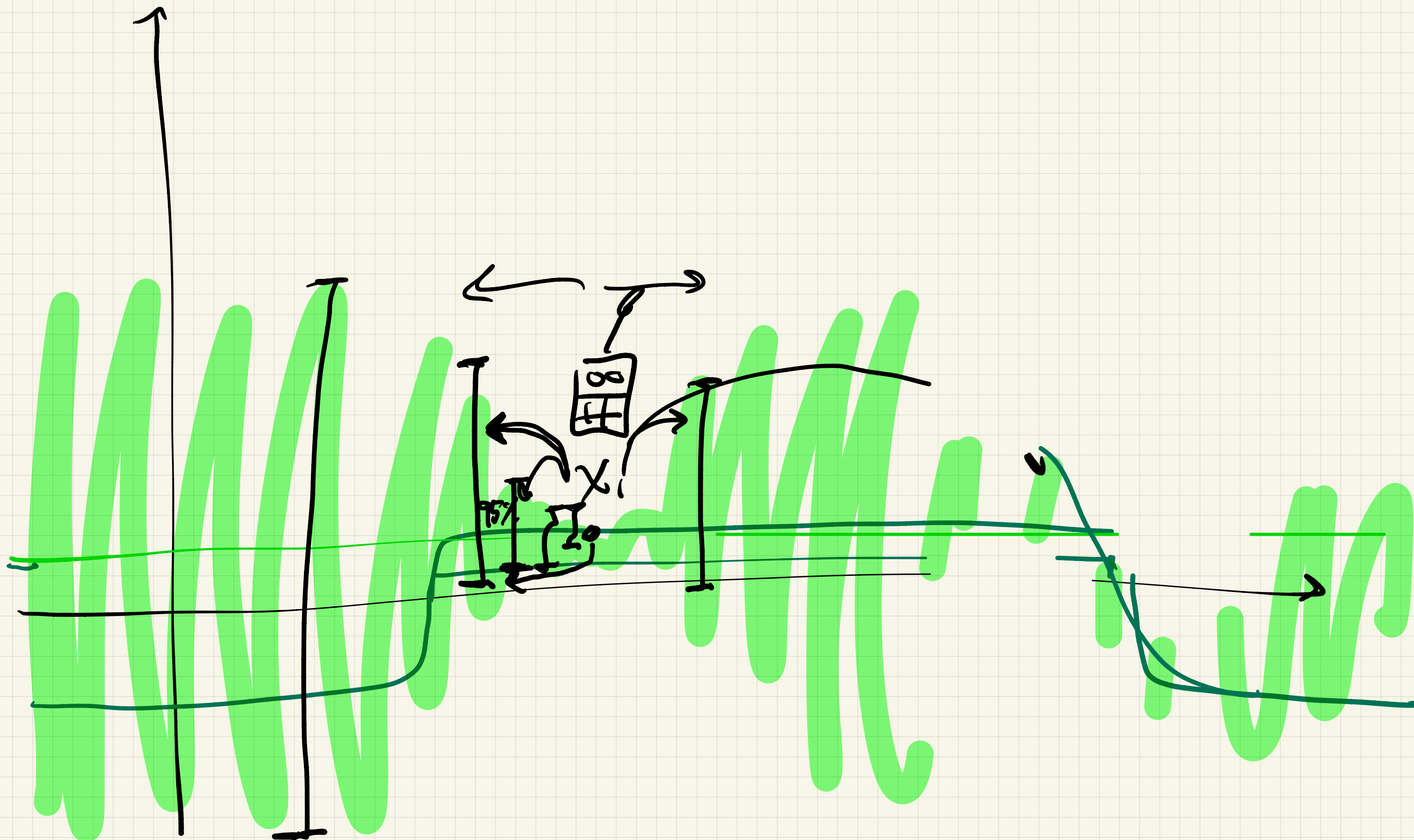


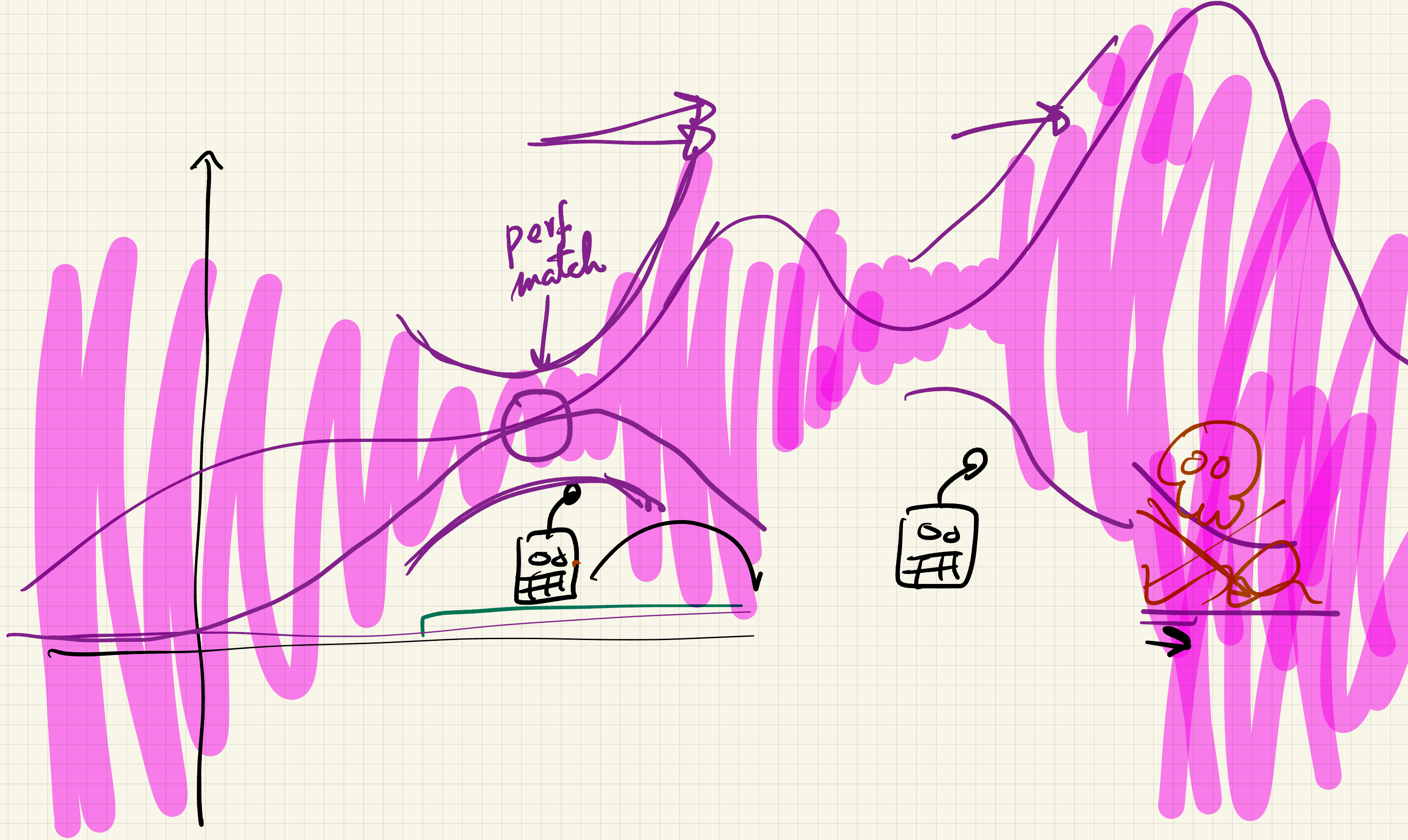
Goal: find $x_n \in \mathcal{S}$ most informative about f on $\mathcal{A}$.

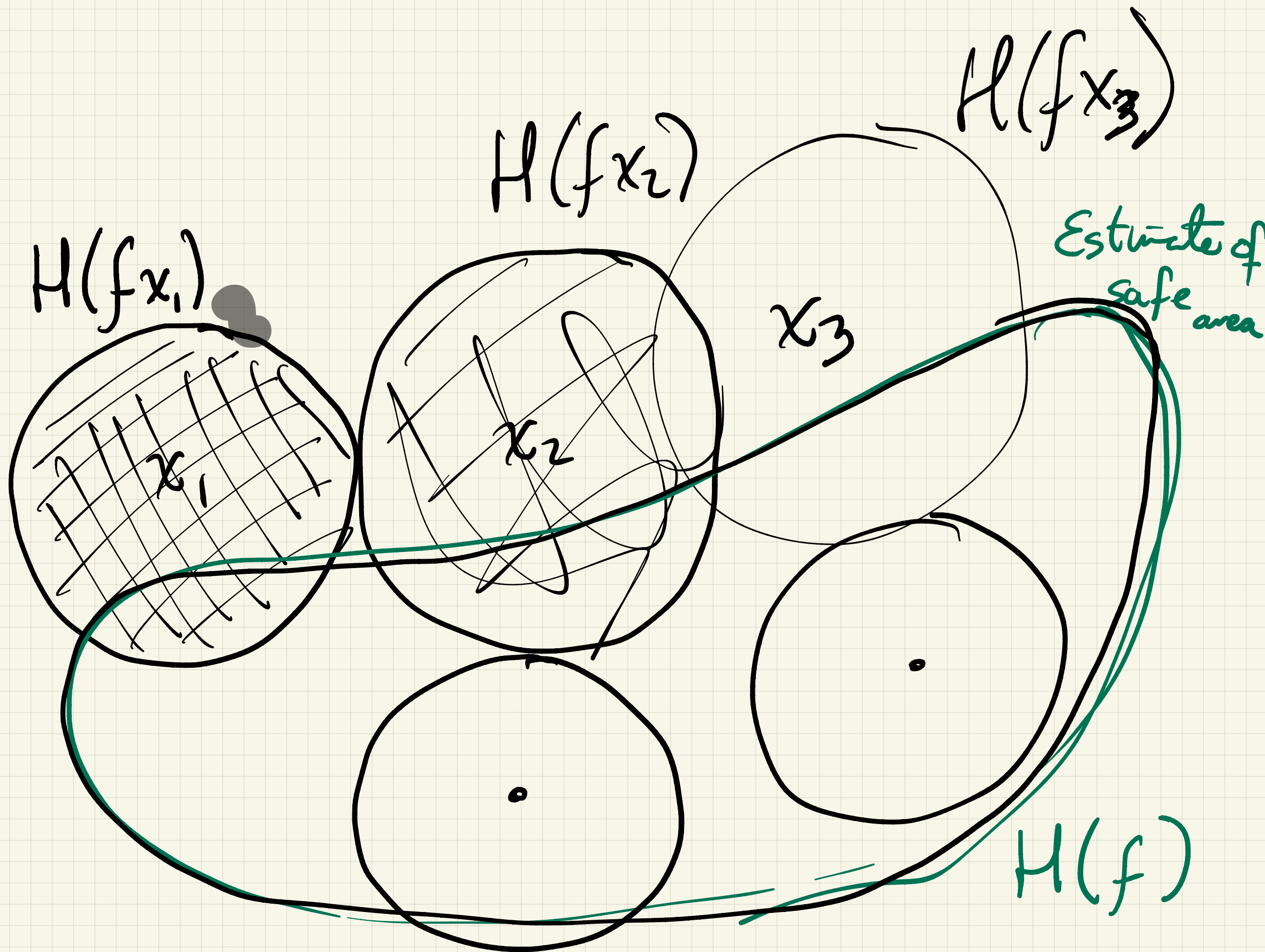
Information-based transductive learning

1D









To pick the next point:

Consider all candidate points $x_1, \dots, x_n$ not studied yet.

For each of them you compute

$$I(f_A; \underline{f(x_i)} | D)$$

your estimate
of function that
counts # of frogs

your estimate of
frogs at
that pos

data collected
so far.



Information-based transductive learning

We pick $x_n \in \mathcal{S}$ to maximize the conditional mutual information between the observation $y_x = f_x + \epsilon_x$ and f restricted to the target $\mathcal{A}$.

$$x_n = \arg \max_{x \in \mathcal{S}} I(f_{\mathcal{A}}; y_x \mid \mathcal{D}_{n-1})$$

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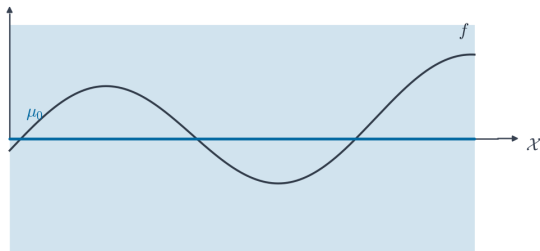
$$I(f_{\mathcal{A}}; y_x \mid \mathcal{D}_{n-1}) = \frac{1}{2} \log \left(\frac{\text{Var}(y_x \mid \mathcal{D}_{n-1})}{\text{Var}(y_x \mid f_{\mathcal{A}}, \mathcal{D}_{n-1})} \right)$$

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The identity holds for $f \sim \mathcal{GP}(\mu, k)$ with known μ, k (numerator: uncertainty in y_x before knowing f on $\mathcal{A}$; denominator: after).

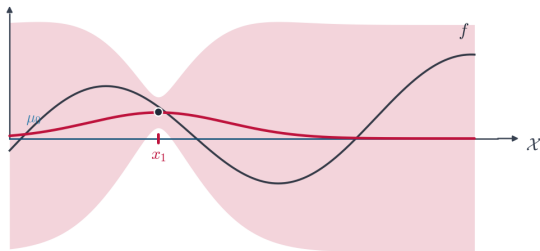
Method: □t and update a GP

- Impose $f \sim \mathcal{GP}(\mu, k)$ on $\mathcal{X}$, with $\mu: \mathcal{X} \rightarrow \mathbb{R}$ and $k: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$.



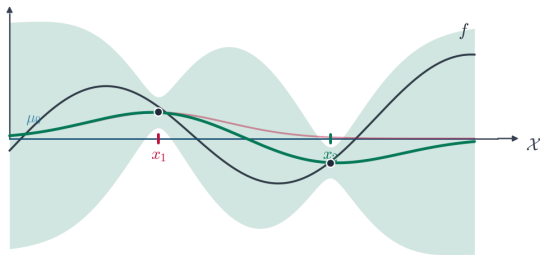
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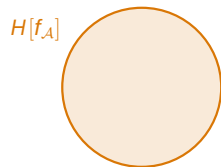


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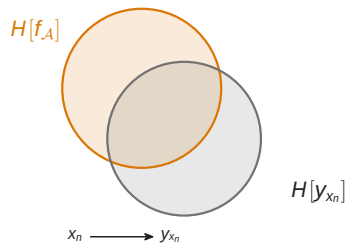
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- For $i = 1, 2, \dots$: pick $x_i \in \mathcal{S}$, observe y_{x_i} , update the posterior with (x_i, y_i) .



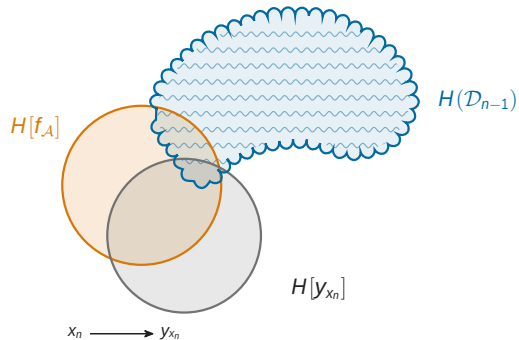
The information-theoretic picture



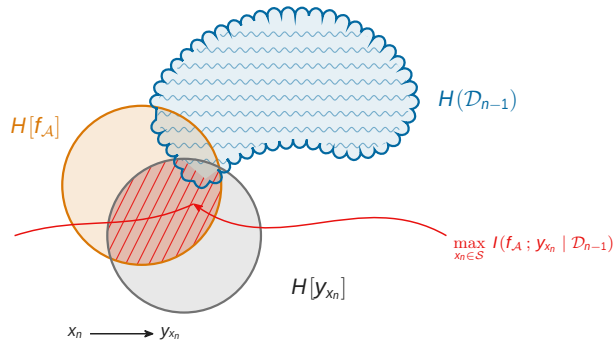
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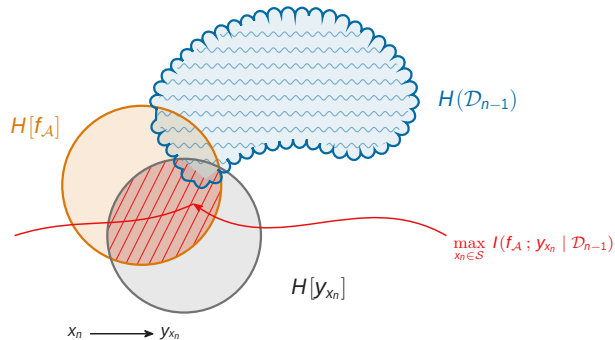
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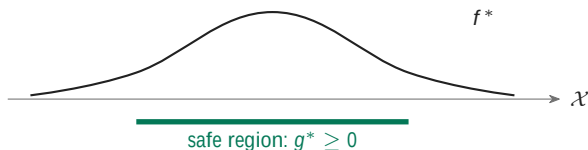
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- Iteratively choose $x_1, \dots, x_{n-1} \in \mathcal{X}$ and observe $y_i = f^*(x_i)$.
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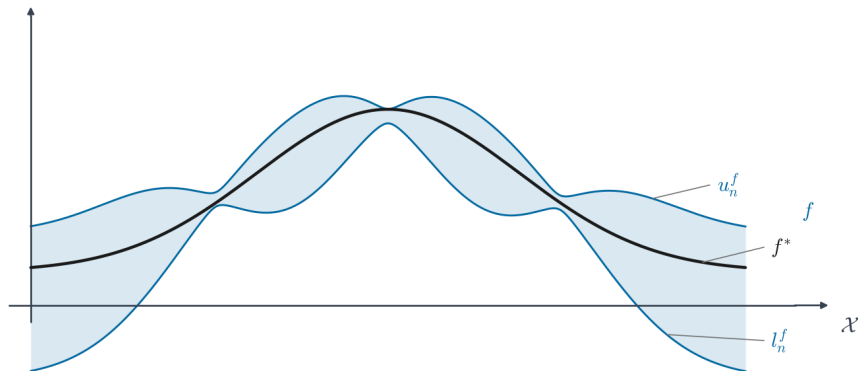
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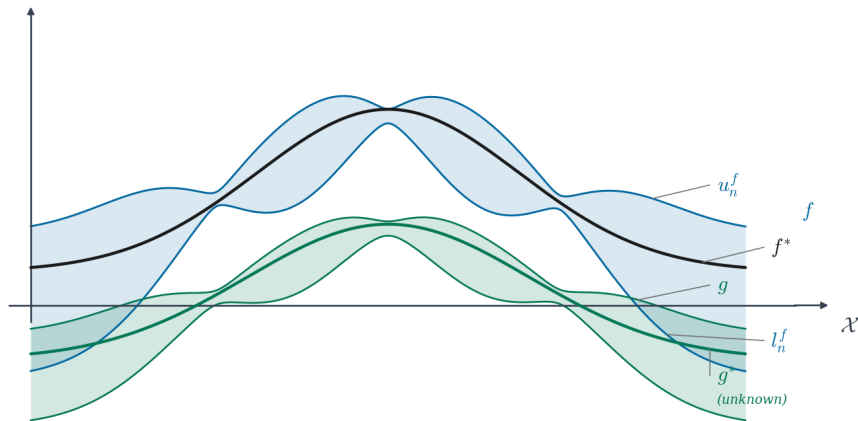
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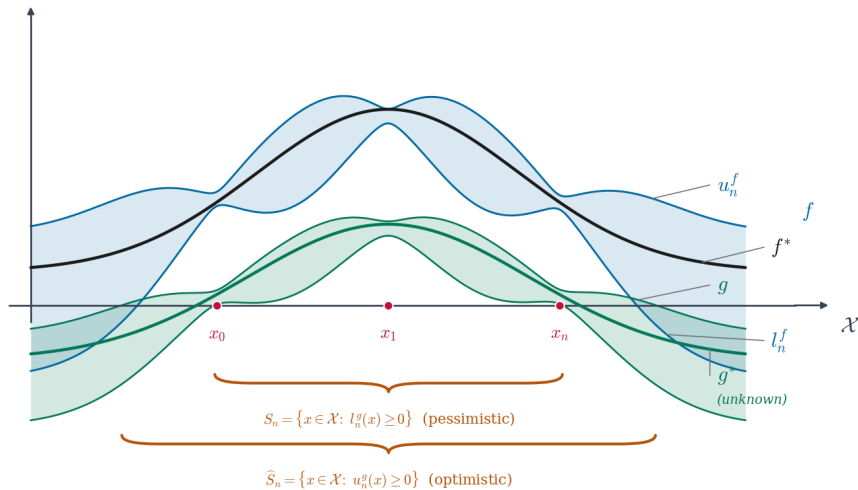
Safe BO: two GPs, two confidence bands



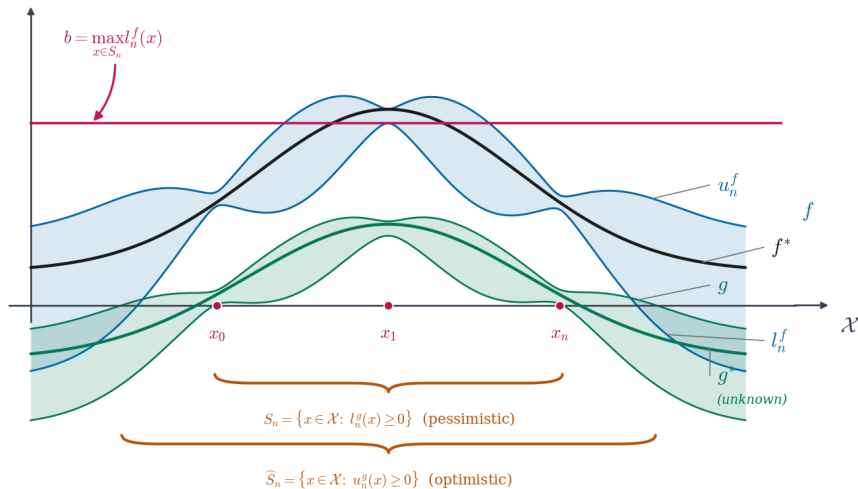
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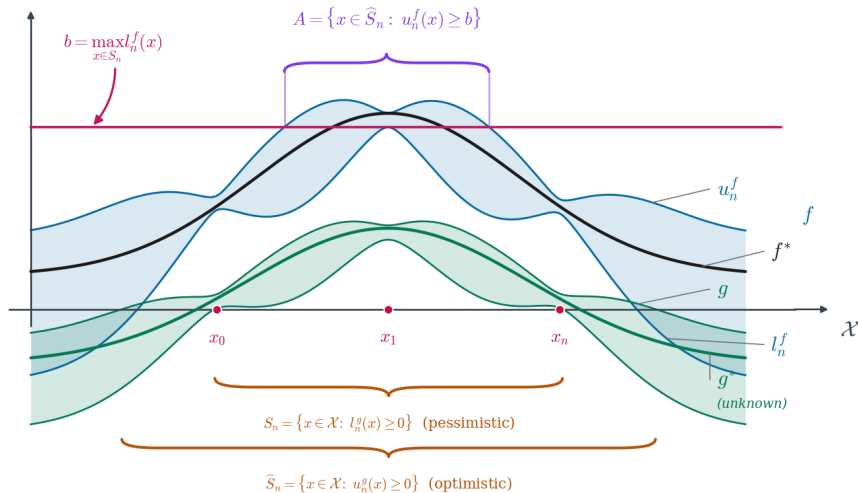
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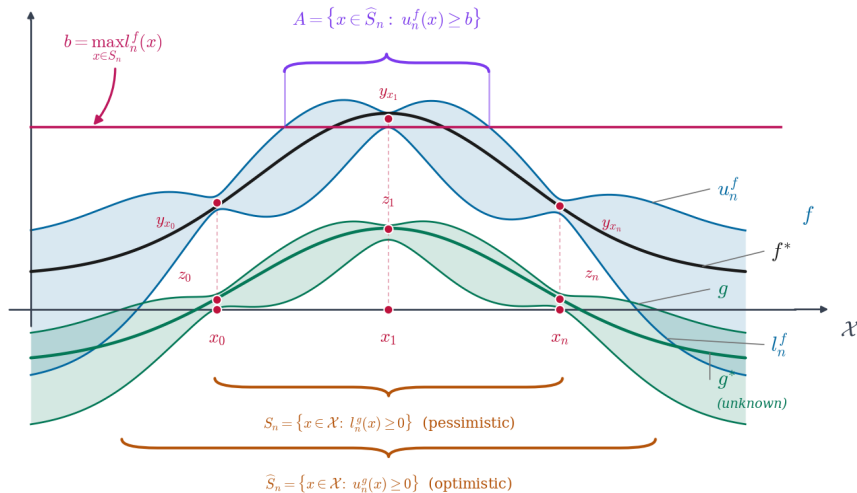
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Safe Bayesian optimization with ITL

- Fit a GP f on $\{(x_i, y_i)\}_{i < n}$: it induces l_n^f, u_n^f with $[l_n^f(x), u_n^f(x)]$ a 95% confidence interval of $\mathbb{E}[f(x)]$.

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- Fit a GP g on $\{(x_i, z_i)\}_{i < n}$: it induces l_n^g, u_n^g , hence safe-set estimates

$$\mathcal{S}_n = \{x : l_n^g(x) \geq 0\} \text{ (pessimistic),} \quad \widehat{\mathcal{S}}_n = \{x : u_n^g(x) \geq 0\} \text{ (optimistic).}$$

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- Then do ITL using sample space $\mathcal{S}_n$ and target space $\mathcal{A}_n$.